Module 06 Lab Reflective Journal

Keira Burdine

In Module 6, I learned more about the difference between regression and classification by working with the Titanic dataset in Google Colab. I used linear regression to predict ticket fares based on passenger age and class, and logistic regression to predict whether a passenger survived based on age, passenger class, and sex. Completing both activities helped me understand that regression predicts numerical values, while classification predicts categories.
I practiced preparing the dataset, selecting features and a target, dividing the data into training and testing sets, and then fitting each model. One important lesson was understanding why we use different evaluation methods. My linear regression model produced a mean squared error of 3,364.92 and a root mean squared error of about $58.01. My logistic regression model had a test accuracy of 74.83%. I learned that the first result measures the size of prediction errors, while accuracy measures the percentage of correct classifications. I also learned to interpret the linear regression coefficients. The coefficient for age was approximately -0.44, and the coefficient for passenger class was approximately -38.30. These values described how the model predicted ticket fares would change when one feature increased while the other stayed the same.
One challenge I faced was making small typing mistakes in Python, including misspelling method and model names. Reading the error messages and correcting my code helped me become more comfortable troubleshooting instead of starting over. I also worked through a Markdown formatting issue in my Knowledge Check responses. Overall, this lab gave me more confidence in using Colab, understanding machine-learning model outputs, and paying attention to details in my code.

John Grubbs

I enjoyed this assignment because it made the difference between regression and classification much clearer through two practical examples. Using the Titanic dataset showed me that regression predicts a continuous numerical value, such as a passenger's fare, while classification predicts a category, such as whether a passenger survived. Seeing both models use related data but produce different kinds of predictions helped the concepts feel more concrete.

I also learned that model evaluation must match the type of problem being solved. Mean squared error was useful for the fare-prediction model because it measures the distance between predicted and actual numerical values. Accuracy was more appropriate for survival prediction because the result is a discrete category that can be counted as correct or incorrect.

The lab also helped me understand that the coefficients in a linear-regression model represent how each feature affects the prediction. Overall, this assignment gave me practice preparing data, training two common machine-learning models, making predictions, and choosing metrics that fit the goal of each model.